ROCSCIENCE

Case2
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:45:25 am
Software Version: 9.041

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Slide2 Analysis Information

Case2

Project Summary

File Name:	Case2.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:00.76s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:45:25 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
Number of slices:	7
Tolerance:	0.005
Maximum number of iterations:	50
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Grid Search
Radius Increment:	10
Composite Surfaces:	Disabled
Reverse Curvature:	Invalid Surfaces
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	20 kN/m3
Cohesion	20 kPa
Phi	20 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	2.112830
Center:	13.689, 25.558
Radius:	15.989
Left Slip Surface Endpoint:	10.001, 10.000
Right Slip Surface Endpoint:	27.499, 17.500
Resisting Moment:	12075.1 kN-m
Driving Moment:	5715.12 kN-m
Total Slice Area:	49.1077 m2
Surface Horizontal Width:	17.4985 m
Surface Average Height:	2.80639 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	1
Number of Invalid Surfaces:	0

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.11283

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.49979	40.9192	-8.80058	Material 1	20	20	12.6223	26.6689	18.3225	0	18.3225	16.3684	16.3684
2	2.49979	112.843	0.219983	Material 1	20	20	17.2309	36.4059	45.0751	0	45.0751	45.1413	45.1413
3	2.49979	164.92	9.24617	Material 1	20	20	20.263	42.8122	62.676	0	62.676	65.9746	65.9746
4	2.49979	196.306	18.5177	Material 1	20	20	21.74	45.933	71.2501	0	71.2501	78.5317	78.5317
5	2.49979	204.159	28.3422	Material 1	20	20	21.5348	45.4994	70.0589	0	70.0589	81.6747	81.6747
6	2.49979	181.923	39.2298	Material 1	20	20	19.2906	40.7577	57.0311	0	57.0311	72.7809	72.7809
7	2.49979	81.0836	52.3794	Material 1	20	20	12.3043	25.9969	16.4763	0	16.4763	32.4419	32.4419

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.11283

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	10.0005	10.0003	0	0	0
2	12.5003	9.61325	38.633	0	0
3	15.0001	9.62285	81.2586	0	0
4	17.4999	10.0298	106.388	0	0
5	19.9997	10.8671	101.058	0	0
6	22.4995	12.2154	60.4058	0	0
7	24.9992	14.2564	-7.78665	0	0
8	27.499	17.5	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	0, 0 35, 0 35, 17.5 25, 17.5 10, 10 0, 10